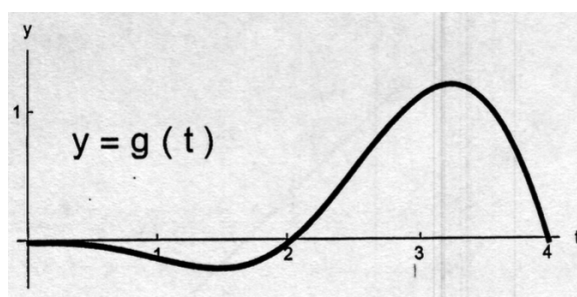


Problem 2

Problem 2

Problem 1 The graph of g , a continuous function on $[0, 4]$, is shown in the figure. Let $A(x) = \int_0^x g(t) dt$, for $0 \leq x \leq 4$.



(a) Circle the correct statement about $A(2)$.

- (i) $A(2) = 0$;
- (ii) $A(2) > 0$;
- (iii) $A(2) < 0$;
- (iv) none of the previous answers.

Explanation. Correct choice: (iii) $A(2) < 0$. The curve $g(t)$ on the interval $(0, 2)$ is below the x -axis and therefore the net area is negative.

(b) Circle the correct statement about $A(3.8)$.

- (i) $A(3.8) = 0$;
- (ii) $A(3.8) > 0$;
- (iii) $A(3.8) < 0$;
- (iv) none of the previous answers.

Explanation. Correct choice: (ii) $A(3.8) > 0$. The net area under the curve $g(t)$ on the interval $(2, 3.8)$ is positive and larger than the negative net area on the interval $(0, 2)$. Therefore, the net area on the interval $(0, 3.8)$ is positive.

(c) Circle the correct statement about $A'(3.8)$.

- (i) $A'(3.8) = 0$;

- (ii) $A'(3.8) > 0$;
- (iii) $A'(3.8) < 0$;
- (iv) none of the previous answers.

Explanation. Correct choice: (ii) $A'(3.8) > 0$ because $A'(3.8) = g(3.8)$

- (d) Find the solution of the following initial value problem: $y'(x) = g(x)$ and $y(0) = 2$.

- (i) $y(x) = g(x)$;
- (ii) $y(x) = g(x) + 2$;
- (iii) $y(x) = A(x)$;
- (iv) $y(x) = A(x) + 2$;
- (v) $y(x) = g'(x)$;
- (vi) $y(x) = g'(x) + 2$;
- (vii) none of the previous answers.

Explanation. Correct choice: (iv) $y(x) = A(x) + 2$. $y'(x) = A'(x) = g(x)$ and $y(0) = A(0) + 2 = 2$

[2]

- (e) Find the expression for $\int_0^4 |g(t)| dt$.

- (i) $A(4)$;
- (ii) $A(2) - A(4)$;
- (iii) $A(4) - A(2)$;
- (iv) $A(4) - 2A(2)$;
- (v) none of the previous answers.

Explanation. Correct choice: (iv) $A(4) - 2A(2)$.

$$\int_0^4 |g(t)| dt = \int_2^4 g(t) dt - \int_0^2 g(t) dt$$

and

$$\int_2^4 g(t) dt = \int_0^4 g(t) dt - \int_0^2 g(t) dt$$

Therefore,

$$\int_0^4 |g(t)| dt = \int_0^4 g(t) dt - 2 \int_0^2 g(t) dt = A(4) - 2A(2)$$

- (f) Find the midpoint Riemann sum for the function g on the interval $[0, 4]$ with $n = 2$ (the number of subintervals).

Problem 2

- (i) $g(1) + g(3)$;
- (ii) $g(0) + g(2)$;
- (iii) $g(2) + g(4)$;
- (iv) $2g(1) + 2g(3)$;
- (v) $2g(0) + 2g(2)$;
- (vi) $2g(2) + 2g(4)$;
- (vii) *none of the previous answers.*

Explanation. Correct choice: (iv) $2g(1) + 2g(3)$

